Central Limit Theorem

Lecture Notes with Step-by-Step Examples

Zamir Hussain
Department of Mathematics
University of Wah

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Abstract

The Central Limit Theorem (CLT) is one of the most powerful results in statistics. It states that the sum (or average) of a large number of independent random variables with finite variance is approximately normally distributed, regardless of the underlying distribution. These notes present the theorem, its statement, and seven fully solved examples illustrating its application to binomial, Poisson, exponential, uniform, and general populations. Each example includes detailed step-by-step calculations and a corresponding graph of the standard normal distribution with the relevant area shaded.

1 Introduction

With the sampling distribution obtained via the CLT we are able to:

- study estimators for large and small samples;
- perform tests of hypothesis for large and small samples.

2 Central Limit Theorem – Statement

Central Limit Theorem (CLT)

Let $X_1, X_2, \ldots, X_n$ be independent random variables with

$$E(X_i) = \mu_i, \qquad \mathrm{Var}(X_i) = \sigma_i^2 < \infty.$$

Define the sum $S_n = X_1 + X_2 + \cdots + X_n$. Then, as $n \to \infty$, the distribution of S_n tends to a normal distribution:

$$S_n \overset{\text{approx.}}{\sim} N(\mu, \sigma^2),$$

where

$$\mu = \sum_{i=1}^{n} \mu_i, \qquad \sigma^2 = \sum_{i=1}^{n} \sigma_i^2.$$

Equivalently, for the sample mean $\bar{X} = S_n/n$,

$$\bar{X} \overset{\text{approx.}}{\sim} N\Big(\mu_{\bar{X}},\ \sigma_{\bar{X}}^2\Big), \quad \mu_{\bar{X}} = \frac{1}{n}\sum_{i=1}^{n} \mu_i, \;\; \sigma_{\bar{X}}^2 = \frac{1}{n^2}\sum_{i=1}^{n} \sigma_i^2.$$

3 Examples

Example 1: Tossing a coin 200 times

A fair coin is tossed 200 times. Find the approximate probability that the number of heads obtained is between 80 and 120.

Step-by-step solution.

1. **Random variable and distribution.** Let X be the number of heads in 200 tosses. $X \sim \text{Binomial}(n = 200,\ p = 0.5)$.

2. **Parameters.**

$$E(X) = np = 200\cdot 0.5 = 100, \qquad \text{Var}(X) = np(1-p) = 200\cdot 0.5\cdot 0.5 = 50, \qquad \sigma_X = \sqrt{50} \approx 7.071.$$

3. **Apply CLT (normal approximation).** Since n is large, $X \overset{\text{approx.}}{\sim} N(\mu = 100,\ \sigma^2 = 50)$.

4. **Standardize.**

$$Z = \frac{X - 100}{\sqrt{50}} \sim N(0,1) \text{ (approx.)}.$$

We need $P(80 < X < 120)$:

$$P\Big(\frac{80-100}{7.071} < Z < \frac{120-100}{7.071}\Big) = P(-2.828 < Z < 2.828).$$

Using the standard normal table (given $P(Z < -2.82) = 0.0024$ and symmetry):

$$P(-2.82 < Z < 2.82) = 1 - 2 \times 0.0024 = 0.9952.$$

5. **Probability.**

$$P(80 < X < 120) \approx 0.9952.$$

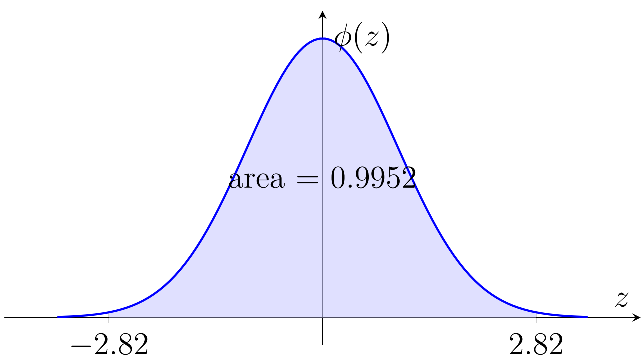

Figure 1: Standard normal curve with shaded region $(-2.82, 2.82)$; probability ≈ 0.9952.

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Given. $n = 100,\ \mu = 60,\ \sigma^2 = 400\ \Rightarrow\ \sigma = 20.$

Sampling distribution of $\bar{X}$. By CLT,

$$E(\bar{X}) = \mu = 60, \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{400}{100} = 4, \qquad \sigma_{\bar{X}} = 2.$$

Hence $\bar{X} \overset{\text{approx.}}{\sim} N(60, 4)$.

Standardize.

$$Z = \frac{\bar{X} - 60}{2} \sim N(0,1).$$

The condition "not differ by more than 4" means $|\bar{X} - 60| \le 4$:

$$P(56 \le \bar{X} \le 64) = P\left(\frac{56-60}{2} \le Z \le \frac{64-60}{2}\right) = P(-2 \le Z \le 2).$$

Normal table. Given $P(Z < -2) = 0.0228$. By symmetry,

$$P(-2 \le Z \le 2) = 1 - 2 \times 0.0228 = 0.9544.$$

(Alternatively, $P(0 < Z < 2) = 0.4772$ gives the same result.)

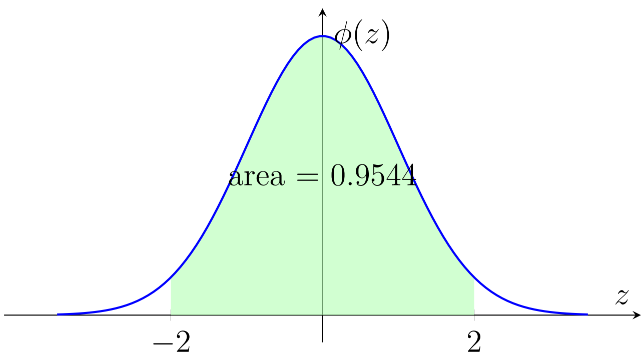

Figure 2: Standard normal curve: shaded area between $z = -2$ and $z = 2$ is 0.9544.

Example 3: Sum of independent Poisson variables

Let $X_1, X_2, \dots, X_n$ be independent Poisson random variables with parameter $\lambda = 2$. Use the CLT to estimate $P(120 \le S_n \le 160)$ where $S_n = X_1 + X_2 + \cdots + X_n$ and $n = 75$.

Step-by-step solution.

1. **Distribution of each X_i.** $X_i \sim \text{Poisson}(\lambda = 2)$. $E(X_i) = 2, \quad \text{Var}(X_i) = 2$.

2. **Mean and variance of the sum.**

$$E(S_n) = n \cdot 2 = 75 \cdot 2 = 150, \qquad \text{Var}(S_n) = n \cdot 2 = 150, \qquad \sigma_{S_n} = \sqrt{150} \approx 12.247.$$

3. **Apply CLT.** $S_n \overset{\text{approx.}}{\sim} N(\mu = 150,\ \sigma^2 = 150)$.

4. **Standardize.**

$$Z = \frac{S_n - 150}{\sqrt{150}} \sim N(0, 1).$$

For $S_n = 120$: $z_1 = \frac{120-150}{12.247} = -2.4495$. For $S_n = 160$: $z_2 = \frac{160-150}{12.247} = 0.8165$.

5. **Use normal table (given).**

$$P(Z < 0.8165) = 0.7939 \implies P(0 < Z < 0.8165) = 0.2939,$$
$$P(Z < -2.45) = 0.0071 \implies P(-2.45 < Z < 0) = 0.4929.$$

(We approximate -2.4495 by -2.45.)

6. **Combine.**

$$P(120 \le S_n \le 160) = P(-2.4495 \le Z \le 0.8165) = 0.4929 + 0.2939 = 0.7868.$$

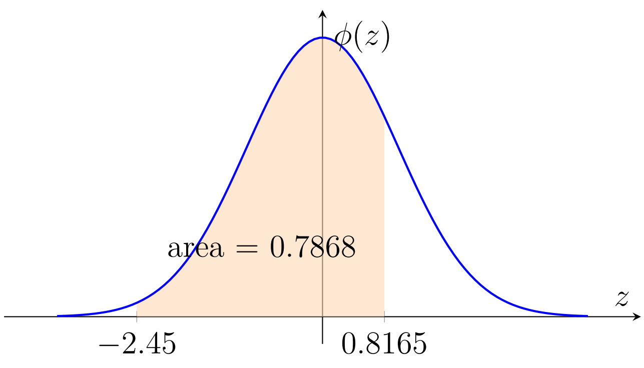

Figure 3: Standard normal curve: shaded area between $z = -2.45$ and $z = 0.8165$ is 0.7868.

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Given. $n = 100,\ \mu = 80,\ \sigma^2 = 400\ \Rightarrow\ \sigma = 20.$

Sampling distribution of $\bar{X}$. By CLT,

$$E(\bar{X}) = 80, \qquad \mathrm{Var}(\bar{X}) = \frac{400}{100} = 4, \qquad \sigma_{\bar{X}} = 2,$$

so $\bar{X} \overset{\text{approx.}}{\sim} N(80, 4)$.

Standardize.

$$Z = \frac{\bar{X} - 80}{2} \sim N(0,1).$$

The condition $|\bar{X} - 80| \le 6$ gives

$$P(74 \le \bar{X} \le 86) = P\left(\frac{74-80}{2} \le Z \le \frac{86-80}{2}\right) = P(-3 \le Z \le 3).$$

Normal table. Given $P(Z > 3) = 0.0014$ (hence $P(Z < -3) = 0.0014$). Therefore,

$$P(-3 \le Z \le 3) = 1 - 2 \times 0.0014 = 0.9972.$$

(Alternatively, $P(0 < Z < 3) = 0.4986$ gives $0.4986 + 0.4986 = 0.9972$.)

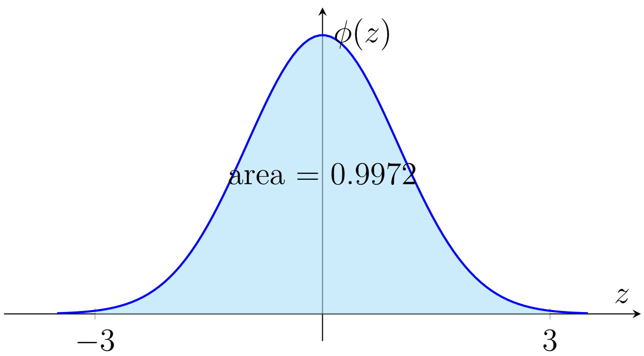

Figure 4: Standard normal curve: shaded area between $z = -3$ and $z = 3$ is 0.9972.

Example 5: Lifetime of electric bulbs (Exponential)

The lifetime of a certain type of electric bulb may be considered as an exponential random variable with mean 50 hours. Using the CLT, find the approximate probability that 100 such bulbs will provide a total of more than 6000 hours of burning time.

Step-by-step solution.

1. **Distribution of each bulb.** Let X_i be the lifetime of the ith bulb. $X_i \sim \text{Exponential}(\lambda)$ with $E(X_i) = 1/\lambda = 50 \Rightarrow \lambda = 1/50$. Variance: $\text{Var}(X_i) = 1/\lambda^2 = 2500$.
2. **Sum of 100 bulbs.** $S_n = X_1 + X_2 + \cdots + X_{100}$.
$$E(S_n) = 100 \cdot 50 = 5000, \qquad \text{Var}(S_n) = 100 \cdot 2500 = 250000, \qquad \sigma_{S_n} = 500.$$
3. **Apply CLT.** $S_n \overset{\text{approx.}}{\sim} N(5000, 250000)$.
4. **Standardize.**
$$Z = \frac{S_n - 5000}{500} \sim N(0,1).$$
We need $P(S_n > 6000) = P\Big(Z > \frac{6000-5000}{500}\Big) = P(Z > 2)$.
5. **Normal table.** Given $P(Z < -2) = 0.0228$; by symmetry $P(Z > 2) = 0.0228$.
6. **Probability.**
$$P(S_n > 6000) \approx 0.0228.$$

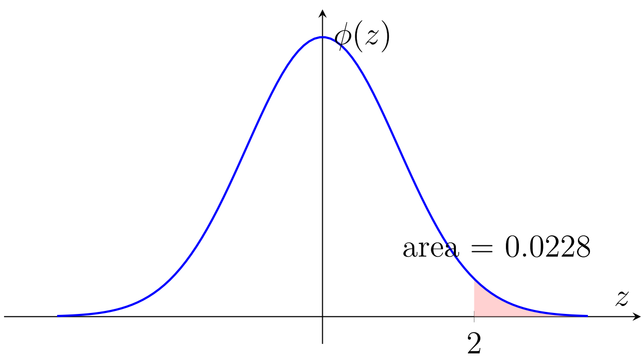

Figure 5: Standard normal curve: right tail area beyond $z = 2$ is 0.0228.

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Given. $n = 50,\ \mu = 100,\ \sigma = 15.$

Sampling distribution of $\bar{X}$. By CLT,

$$E(\bar{X}) = \mu = 100, \qquad \text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{225}{50} = 4.5, \qquad \sigma_{\bar{X}} = \sqrt{4.5} \approx 2.1213.$$

Hence $\bar{X} \overset{\text{approx.}}{\sim} N(100,\ 4.5)$.

Standardize.

$$Z = \frac{\bar{X} - 100}{2.1213} \sim N(0, 1).$$

We need $P(95 < \bar{X} < 105)$:

$$P\left(\frac{95 - 100}{2.1213} < Z < \frac{105 - 100}{2.1213}\right) = P(-2.357 < Z < 2.357).$$

Normal table (approximate). From standard normal tables, $P(0 < Z < 2.36) \approx 0.4909$ (using $z = 2.36$). By symmetry, $P(-2.36 < Z < 2.36) = 2 \times 0.4909 = 0.9818$. (We use 2.36 as a close approximation to 2.357.)

Probability.

$$P(95 < \bar{X} < 105) \approx 0.9818.$$

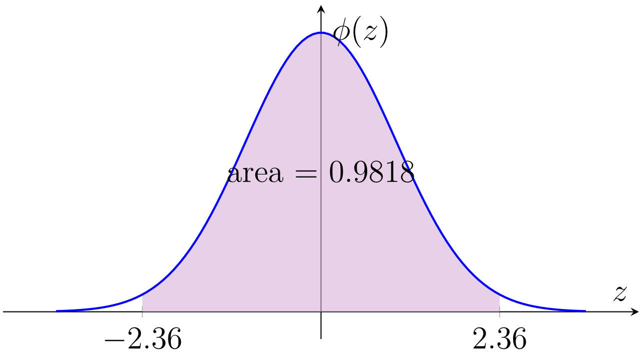

Figure 6: Standard normal curve: shaded area between $z = -2.36$ and $z = 2.36$ is 0.9818.

Example 7: Sum of independent uniform variables

The weight of a randomly selected package is uniformly distributed between 10 kg and 20 kg. If 30 packages are randomly selected, approximate the probability that the total weight exceeds 500 kg. Use the CLT.

Step-by-step solution.

1. **Distribution of each package weight.** Let $X_i \sim \text{Uniform}(10, 20)$.

$$E(X_i) = \frac{10 + 20}{2} = 15, \qquad \text{Var}(X_i) = \frac{(20 - 10)^2}{12} = \frac{100}{12} \approx 8.3333, \qquad \sigma_{X_i} = \sqrt{8.3333} \approx 2.8868.$$

2. **Sum of 30 packages.** $S_n = X_1 + X_2 + \cdots + X_{30}$.

$$E(S_n) = 30 \cdot 15 = 450, \qquad \text{Var}(S_n) = 30 \cdot 8.3333 = 250, \qquad \sigma_{S_n} = \sqrt{250} \approx 15.8114.$$

3. **Apply CLT.** $S_n \overset{\text{approx.}}{\sim} N(450,\ 250)$.

4. **Standardize.**

$$Z = \frac{S_n - 450}{15.8114} \sim N(0, 1).$$

We need $P(S_n > 500) = P\left(Z > \frac{500-450}{15.8114}\right) = P(Z > 3.1623)$.

5. **Normal table.** From standard normal tables, $P(Z < 3.16) \approx 0.9992$. Therefore $P(Z > 3.16) = 1 - 0.9992 = 0.0008$. (Using $z = 3.16$ as an approximation to 3.1623.)

6. **Probability.**
$$P(S_n > 500) \approx 0.0008.$$

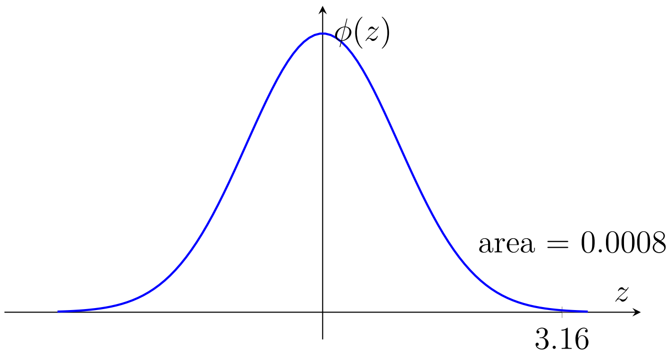

Figure 7: Standard normal curve: right tail area beyond $z = 3.16$ is approximately 0.0008.

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4 Summary of Results

Example **Approx. Probability**	**Distribution**	**Sample size n**	**Event**
1 0.9952	Binomial	200	$80 < X < 120$
2 0.9544	Unknown	100	$\lvert\bar{X} - 60\rvert \le 4$
3 0.7868	Poisson	75	$120 \le S_n \le 160$
4 0.9972	Unknown	100	$\lvert\bar{X} - 80\rvert \le 6$
5 0.0228	Exponential	100	$S_n > 6000$
6 0.9818	Normal*	50	$95 < \bar{X} < 105$
7 0.0008	Uniform	30	$S_n > 500$

Table 1: Overview of the seven CLT examples. *Although the underlying population is normal here, the CLT justifies the use of the normal distribution for $\bar{X}$ even if the population were non-normal; the result is exact in this particular case.

Conclusion

The Central Limit Theorem provides a powerful tool for approximating probabilities associated with sums and averages, even when the underlying population distribution is not normal. The seven examples above demonstrate the standard workflow:

1. Identify the mean and variance of the original random variables.
2. Compute the mean and variance of the sum (or sample mean).
3. Apply the CLT to justify a normal approximation.
4. Standardize to obtain a z-score.
5. Use the standard normal table to find the desired probability.
6. (Optional) Visualise the probability as an area under the normal curve.

All calculations used standard normal tail probabilities; in practice one would consult more precise tables or software, but the CLT approximation remains valid for sufficiently large n.